DEPARTMENT OF CHEMICAL ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY SRINAGAR

Bioresource Technology (CET-465)

Home Assignment plus Question Bank-6

S. No.	Questions	COs
1.	Write a brief note on bioethanol and discuss the process for 2nd generation bioethanol production.	CO3
2.	Briefly explain the global scenario of biofuel production.	CO2
3.	Classify and describe bioethanol based on feedstocks.	CO1
4.	Write short note on the important feedstocks suitable for production of bioethanol.	CO1
5.	Discuss the common flow diagram for the production of bioethanol from the 1^{st} and the 2^{nd} generation of feedstocks.	CO3
6.	Explain the simplified scheme for bioethanol production from sugarcane.	CO3
7.	Make a proper comparison of specifications between gasoline and bioethanol.	CO1
8.	Discuss India's status in bioethanol production, consumption and requirement.	CO1
9.	Write a brief note on bio-aviation turbine fuel and highlight its significance w.r.t. its necessity.	CO1
10.	Enlist the various production technologies used for bioATF and give a diagram explaining production process-wise raw material and technology overview for bio-jet fuel.	CO2
11.	How can the biojet fuel be manufactured from bio-ethanol, explain with the help of a simplified flow diagram.	CO3
12.	What is biobutanol? Discuss its significance as a biofuel.	CO1
13.	Tabulate the structures, properties and main applications of N-butanol, 2-butanol, iso-butanol and tert-butanol.	CO1
14.	Compare the properties of butanol and other fuels.	CO1
15.	Draw a representative schematic diagram for fermentative production of butanol from lignocellulosic biomass.	CO4
16.	Discuss the challenges and their solutions with respect to butanol production.	CO4
17.	Write a brief note on the global scenario of bio-butanol market - growth, trends, and forecast (2019 - 2024).	CO4
CO1:	Fundamental understanding of the bioresources and its applications for attainment of social objectives (energy, environment, product, sustainability).	
CO2:	Acquire knowledge with respect to the properties of the bioresources and the conversion technologies.	
CO3:	Exhibiting knowledge of the systems used for bioresources and bioresource technology.	
CO4:	Understanding about analysis of data and their applications in design of the systems and development of the bioprocess.	